AvenueMart Retail Pvt. Ltd.

Inventory Optimization Analysis

Stock availability, excess inventory, supplier reliability and inventory losses

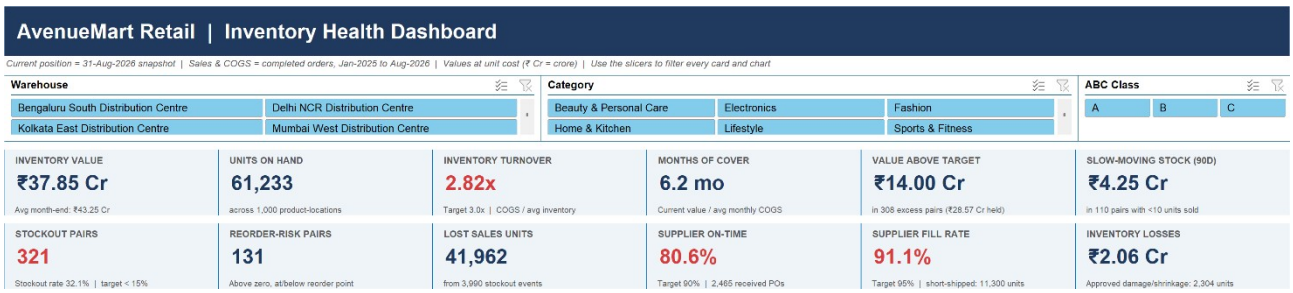


Figure 1: Headline KPIs from the Excel Inventory Health Dashboard (31-Aug-2026)

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Data period: 1 January 2025 to 31 August 2026 (current position = 31-Aug-2026 month-end snapshot)

Analysis workbook: AvenueMart_Inventory_Optimization_Analysis.xlsx (Power Query, Data Model, DAX, PivotTables, Dashboard)

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1. Executive summary

AvenueMart Retail Pvt. Ltd. asked InsightEdge Consulting to review inventory availability, excess stock, replenishment performance and inventory losses across its four regional distribution centres. We cleaned and modelled 20 months of ERP data (250 products, 40 suppliers, 4 warehouses, 1-Jan-2025 to 31-Aug-2026) in Excel using Power Query, the Excel Data Model and DAX. Every headline figure in this report reconciles exactly to the client reference snapshot.

Key messages

- **The stock is in the wrong places.** AvenueMart holds ₹37.85 Cr of inventory (61,233 units), yet 308 product-locations are overstocked while 321 are out of stock with customers waiting.
- **Inventory was cut, but not where it mattered.** Month-end inventory value fell 37% (from ₹59.66 Cr in Jan-2025 to ₹37.85 Cr in Aug-2026) while stockout product-locations rose from 0 to 321.
- **₹18 Cr of cash can be released.** ₹14.00 Cr is held above target stock levels and a further ₹4.25 Cr sits in 110 slow-moving product-locations.
- **Suppliers are a root cause of stockouts.** Only 80.6% of purchase orders arrive on time and 91.1% of ordered quantity is delivered; 41,962 units of demand were lost to stockouts.
- **Losses are a handling problem.** ₹2.06 Cr of approved losses, 78% from damage and shrinkage, concentrated in high-value Electronics and the Delhi NCR warehouse.

What we recommend

Rebalance excess stock between warehouses into stockout locations before buying more; replenish the 452 at-risk product-locations starting with A-class items; freeze purchasing for items above target; clear slow movers through targeted promotions; put the weakest suppliers on improvement plans; and tighten handling controls for high-value Electronics. Section 5 sets out the prioritised plan and Section 6 the targets to track.

2. Background and objectives

AvenueMart is an Indian omnichannel retailer selling electronics, fashion, home and kitchen, beauty, sports and lifestyle products through stores, its website and a mobile app. It operates four regional distribution and fulfilment centres (Delhi NCR, Mumbai, Bengaluru and Kolkata). Management has sales and replenishment data but no clear view of where inventory is too high, where it is running out, which suppliers are reliable and where stock is being lost.

Management questions

1. How much inventory do we currently hold, and which warehouses carry the most value?
2. Which categories and products carry high inventory value relative to their sales contribution?
3. Which product-location combinations are at stockout or reorder risk?
4. Which products look slow-moving or excessively stocked based on recent demand?
5. Which suppliers are reliable on delivery timing and quantity fulfilment?
6. Where are inventory losses and stock adjustments occurring?
7. What should AvenueMart do next to reduce excess inventory and stockouts without disrupting sales?

Scope

The analysis covers 20 month-end inventory snapshots (20,000 product-location-month records), 18,233 sales lines, 2,475 purchase orders, 1,191 stock adjustments and 3,990 stockout events. "Current" always means the 31-Aug-2026 snapshot. Sales, COGS and profit use completed orders only; cancelled orders are excluded from demand and COGS.

3. Data and approach

3.1 Data sources and cleaning results

Table 1: Raw versus clean row counts after de-duplication (Power Query)

Table (query)	Raw rows	Duplicates removed	Clean rows
dim_Products	251	1	250
dim_Suppliers	41	1	40
dim_Warehouses	5	1	4
dim_Calendar	608	0	608
fact_Inventory	20,001	1	20,000
fact_Sales	18,234	1	18,233
fact_PurchaseOrders	2,476	1	2,475
fact_StockAdjustments	1,192	1	1,191
fact_StockoutEvents	3,991	1	3,990

3.2 Data preparation in Power Query

All nine raw files are connected to the workbook through Power Query, so the analysis refreshes with one click (Data > Refresh All). The intentionally messy extracts were cleaned with reusable functions:

- **Standardised IDs and text:** removed stray spaces and upper-cased all IDs (e.g. "prd0003" to "PRD0003"); applied consistent casing to categories, brands, channels and statuses.
- **Parsed numbers and dates:** removed ₹ symbols, spaces and thousand separators; converted three discount formats (10%, 10 and 0.10) to one; typed all dates.
- **Split combined fields:** warehouse|product keys, stock policy (reorder point | safety stock | target stock), the 7-part stock flow, supplier profile, warehouse location and stockout details.
- **Removed duplicates at the correct grain:** one duplicate row in every source; inventory de-duplicated by snapshot date x warehouse x product.
- **Derived business fields:** inventory value at unit cost, stock status (Stockout, No Demand / Zero Stock, Reorder Risk, Excess, Healthy), value above target, PO on-time flag, delay days, short-shipped quantity and adjustment value.

3.3 Data model and measures

The cleaned tables form a star schema in the Excel Data Model: four dimensions (Products, Warehouses, Suppliers, Calendar) and five fact tables (Inventory snapshots, Sales, Purchase orders, Stock adjustments, Stockout events) joined by 16 one-to-many relationships. 56 DAX measures calculate every KPI, including latest-snapshot measures that always read the most recent month in the current filter. Appendix C lists the key formulas.

3.4 Validation

Referential integrity was checked (every product, warehouse and supplier in the fact tables exists in the dimensions; all dates fall inside the calendar). Inventory arithmetic (opening + received + adjustment - sold = closing) holds for 19,837 of 20,000 rows; the remaining 163 are negative adjustments on stock that was already zero, so closing stock was floored at zero. These rows were flagged, not altered. All 18 headline KPIs match the client reference snapshot:

Table 2: KPI reconciliation - Excel Data Model vs. client reference snapshot

KPI	Data Model value	Reference (brief)	Check
Current inventory units	61,233	61,233	Match
Current inventory value	₹37.85 Cr	₹37.85 Cr	Match
Completed sales	₹158.65 Cr	₹158.65 Cr	Match
Completed COGS	₹122.14 Cr	₹122.14 Cr	Match
Completed profit	₹36.52 Cr	₹36.52 Cr	Match
Inventory turnover	2.82x	2.82x	Match
Current stockout pairs	321	321	Match
Current reorder-risk pairs	131	131	Match
Current excess pairs	308	308	Match
Current excess inventory value	₹28.57 Cr	₹28.57 Cr	Match
Above-target excess value	₹14.00 Cr	₹14.00 Cr	Match
Stockout events	3,990	3,990	Match
Lost sales units	41,962	41,962	Match
Supplier on-time rate	80.6%	80.6%	Match
Supplier fill rate	91.1%	91.1%	Match
Slow-moving pairs (90 days)	110	110	Match
Slow-moving inventory value	₹4.25 Cr	₹4.25 Cr	Match
Approved negative adjustment value	₹2.06 Cr	₹2.06 Cr	Match

3.5 Assumptions and limitations

- Inventory is valued at each product's unit cost from the product master; no write-down or market valuation is applied.
- The supplier profile field was read as Region | Lead time (days) | Payment terms (days); it is descriptive only and not used in any KPI.
- Lost sales value is the estimate recorded on stockout events and is indicative rather than audited.
- Slow-moving uses the brief's rule: positive stock with fewer than 10 completed units sold in the 90 days to 31-Aug-2026.

4. Findings

4.1 How much inventory do we hold, and where? (Q1)

AvenueMart holds 61,233 units worth ₹37.85 Cr at cost. Kolkata East carries the most value (₹12.19 Cr, 32%) and the most units, and it also has the slowest turnover (2.17x against 2.82x overall). Delhi NCR holds the least (₹7.16 Cr) and turns fastest (3.44x). No warehouse is short of space (31-55% of capacity used), so the issue is how stock is allocated, not storage capacity. Electronics alone accounts for ₹23.58 Cr (62%) of the value.

Table 3: Current inventory position by warehouse (31-Aug-2026)

Warehouse_Name	Units on Hand	Inventory Value (₹ Cr)	Value Share %	Capacity Used %	Turnover (x)
Kolkata East Distribution Centre	17,741	₹12.19 Cr	32.2%	55.4%	2.17
Bengaluru South Distribution Centre	15,291	₹10.42 Cr	27.5%	42.5%	2.72
Mumbai West Distribution Centre	15,130	₹8.08 Cr	21.4%	39.8%	3.26
Delhi NCR Distribution Centre	13,071	₹7.16 Cr	18.9%	31.1%	3.44
Grand Total	**61,233**	**₹37.85 Cr**	**100.0%**	**41.4%**	**2.82**

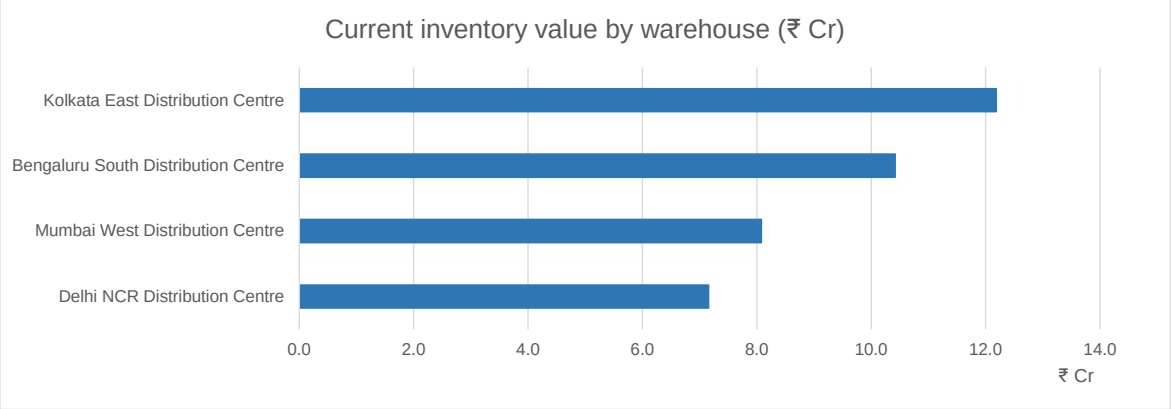


Figure 2: Current inventory value by warehouse (₹ Cr) - Excel Dashboard

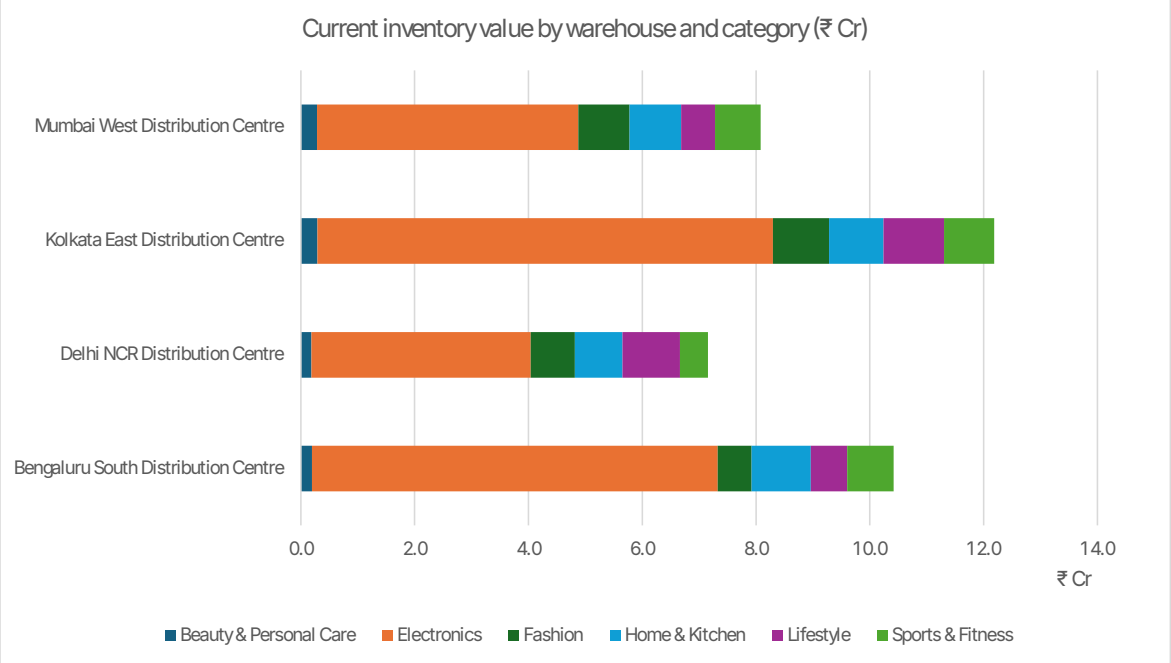


Figure 3: Current inventory value by warehouse and category - Excel sheet Q1

4.2 Which categories and products are over-invested? (Q2)

Electronics is the engine of the business: it earns 77.7% of completed sales with 62.3% of inventory value. Every other category holds a bigger share of inventory than of sales. Sports & Fitness (ratio 1.94), Home & Kitchen (1.89) and Fashion (1.82) carry roughly 11-12 months of cover against 6.2 months for the company. A small number of Electronics products hold more than a year of stock: the IdeaPad Slim 5 (₹3.88 Cr, 21 months of cover), the 43-inch 4K UHD Smart TV (₹2.11 Cr, 24 months) and the Galaxy M35 5G (₹1.91 Cr, 17 months).

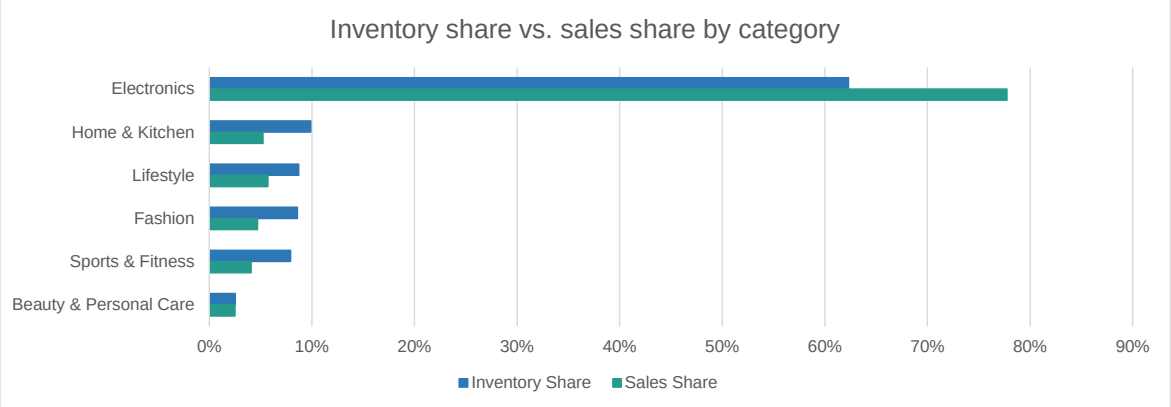


Figure 4: Inventory value share vs. completed sales share by category

Table 4: Category inventory share vs. sales share (sorted by inventory-to-sales ratio)

Category	Inventory Value (₹ Cr)	Inventory Share %	Completed Sales (₹ Cr)	Sales Share %	Inv/Sales Share Ratio	Turnover (x)	Months of Cover
Sports & Fitness	₹3.00 Cr	7.9%	₹6.49 Cr	4.1%	1.94	1.78	12.2
Home & Kitchen	₹3.75 Cr	9.9%	₹8.31 Cr	5.2%	1.89	1.70	12.2
Fashion	₹3.25 Cr	8.6%	₹7.47 Cr	4.7%	1.82	1.85	11.2
Lifestyle	₹3.30 Cr	8.7%	₹9.06 Cr	5.7%	1.53	2.02	9.8
Beauty & Personal Care	₹0.96 Cr	2.5%	₹3.97 Cr	2.5%	1.02	2.71	6.1
Electronics	₹23.58 Cr	62.3%	₹123.34 Cr	77.7%	0.80	3.26	4.9
Grand Total	**₹37.85 Cr**	**100.0%**	**₹158.65 Cr**	**100.0%**	**1.00**	**2.82**	**6.2**

Table 5: Top 10 products by current inventory value (all warehouses)

Product_Label	Category	Inventory Value (₹ Cr)	Completed Sales (₹ Cr)	Months of Cover	Units Sold 90D
PRD0018 - IdeaPad Slim 5	Electronics	₹3.88 Cr	₹5.47 Cr	21.0	117
PRD0031 - 43-inch 4K UHD Smart TV	Electronics	₹2.11 Cr	₹2.13 Cr	24.3	52
PRD0003 - Galaxy M35 5G	Electronics	₹1.91 Cr	₹2.66 Cr	17.3	63
PRD0039 - 55-inch Bravia 3	Electronics	₹1.78 Cr	₹5.58 Cr	9.6	77
PRD0014 - Inspiron 14	Electronics	₹1.14 Cr	₹7.05 Cr	3.7	84
PRD0037 - 43-inch Bravia 3	Electronics	₹1.08 Cr	₹4.77 Cr	5.2	70
PRD0035 - 50-inch Crystal 4K TV	Electronics	₹1.05 Cr	₹3.39 Cr	7.4	47
PRD0009 - 13R	Electronics	₹0.85 Cr	₹2.42 Cr	8.8	76
PRD0038 - 50-inch Bravia 3	Electronics	₹0.80 Cr	₹5.34 Cr	3.8	77
PRD0001 - Galaxy A25 5G	Electronics	₹0.71 Cr	₹1.38 Cr	11.8	62

4.3 Where are we at stockout or reorder risk? (Q3)

Stockouts have become the norm rather than the exception. In January 2025 no product-location was out of stock; by August 2026, 321 of 1,000 were (32.1%), peaking at 342 in June 2026. Over the same period inventory value fell by ₹21.8 Cr. A further 131 product-locations are at or below their reorder point, so 452 (45%) need replenishment now, including 79 A-class (fastest-moving) pairs. Delhi NCR has the highest stockout rate (39.6%) followed by Mumbai West (35.2%). Over the 20 months there were 3,990 stockout

events and 41,962 units of unfulfilled demand. The full product-level action list is in the workbook (sheet Q3, Table 3.4).

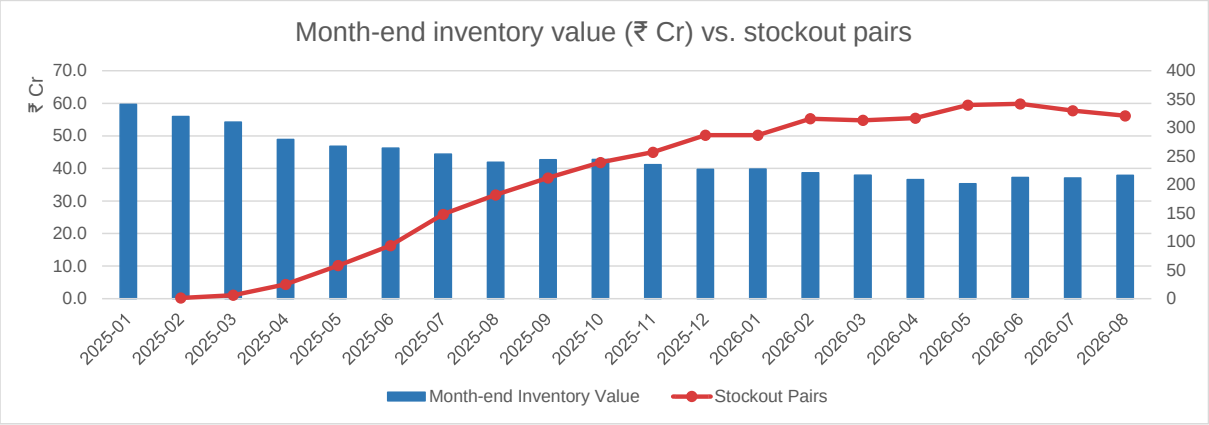


Figure 5: Month-end inventory value (₹ Cr, columns) vs. product-locations in stockout (line)

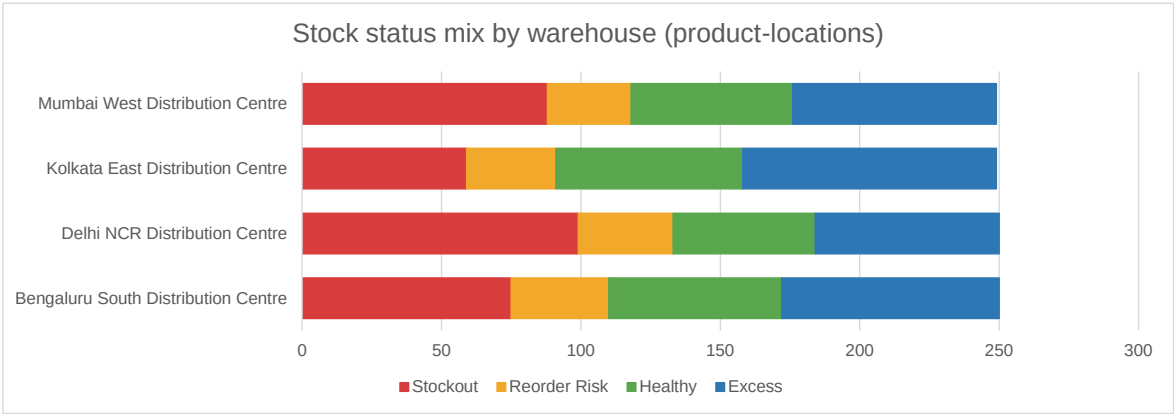


Figure 6: Current stock status mix by warehouse (number of product-locations)

Table 6: Current stock status by warehouse (product-location pairs)

Warehouse_Name	Total Pairs	Stockout	Reorder Risk	Healthy	Excess	No Demand / Zero Stock	Stockout Rate %
Bengaluru South Distribution Centre	250	75	35	62	78		30.0%
Delhi NCR Distribution Centre	250	99	34	51	66		39.6%
Kolkata East Distribution Centre	250	59	32	67	91	1	23.6%
Mumbai West Distribution Centre	250	88	30	58	73	1	35.2%
Grand Total	**1,000**	**321**	**131**	**238**	**308**	**2**	**32.1%**

Table 7: Current risk and stockout history by category

Category	Stockout Pairs	Reorder-Risk Pairs	Stockout Events (history)	Lost Sales Units	Lost Sales Value (₹ Cr)
Electronics	86	27	1,107	13,354	₹61.55 Cr
Beauty & Personal Care	63	23	684	7,517	₹1.11 Cr
Fashion	49	20	684	7,745	₹3.38 Cr
Lifestyle	43	19	555	4,974	₹2.42 Cr
Sports & Fitness	42	27	518	4,301	₹1.24 Cr
Home & Kitchen	38	15	442	4,071	₹1.55 Cr
Grand Total	**321**	**131**	**3,990**	**41,962**	**₹71.25 Cr**

4.4 Which products are slow-moving or overstocked? (Q4)

308 product-locations are above their target stock and hold ₹28.57 Cr; the units above target alone are worth ₹14.00 Cr (25,683 units). Half of that (₹7.15 Cr) sits in just 10 products, led by the IdeaPad Slim 5 (₹2.17 Cr above target across warehouses; 395 units against a target of 77 at Bengaluru). Separately, 110 product-locations with stock on hand sold fewer than 10 units in the last 90 days, tying up ₹4.25 Cr. Fashion has the most slow-moving pairs (33) and Kolkata the most slow-moving value (₹2.26 Cr).

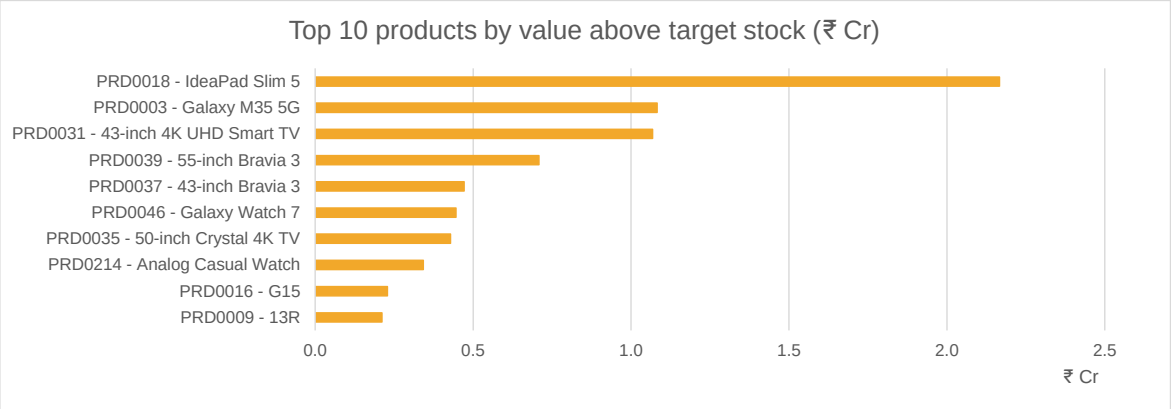


Figure 7: Top 10 products by value held above target stock (₹ Cr)

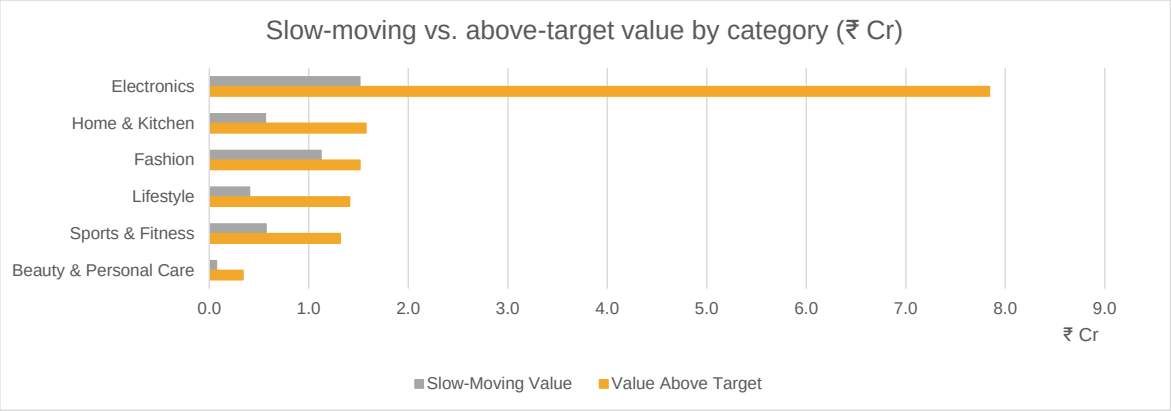


Figure 8: Slow-moving value vs. value above target, by category (₹ Cr)

Table 8: Slow-moving and excess stock by category

Category	Slow-Moving Pairs	Slow-Moving Value (₹ Cr)	Excess Pairs	Excess Pair Value (₹ Cr)	Value Above Target (₹ Cr)	Months of Cover
Electronics	10	₹1.51 Cr	45	₹17.10 Cr	₹7.84 Cr	4.9
Home & Kitchen	20	₹0.57 Cr	62	₹2.95 Cr	₹1.57 Cr	12.2
Fashion	33	₹1.12 Cr	57	₹2.76 Cr	₹1.51 Cr	11.2
Lifestyle	22	₹0.41 Cr	53	₹2.61 Cr	₹1.41 Cr	9.8
Sports & Fitness	19	₹0.57 Cr	57	₹2.46 Cr	₹1.32 Cr	12.2
Beauty & Personal Care	6	₹0.07 Cr	34	₹0.69 Cr	₹0.34 Cr	6.1
Grand Total	**110**	**₹4.25 Cr**	**308**	**₹28.57 Cr**	**₹14.00 Cr**	**6.2**

4.5 Which suppliers are reliable? (Q5)

Supplier service is adequate on quantity but weak on timing. Across 2,465 received purchase orders, 80.6% arrived by the expected date and late orders were on average 4.6 days late. Fill rates are consistent (89.5-93.1% by supplier, 91.1% overall), but 11,300 ordered units were never delivered. Timing varies widely: five suppliers are at or below 75% on-time, while the best four exceed 87%. Bengaluru South receives the least punctual deliveries (77.8% on-time).

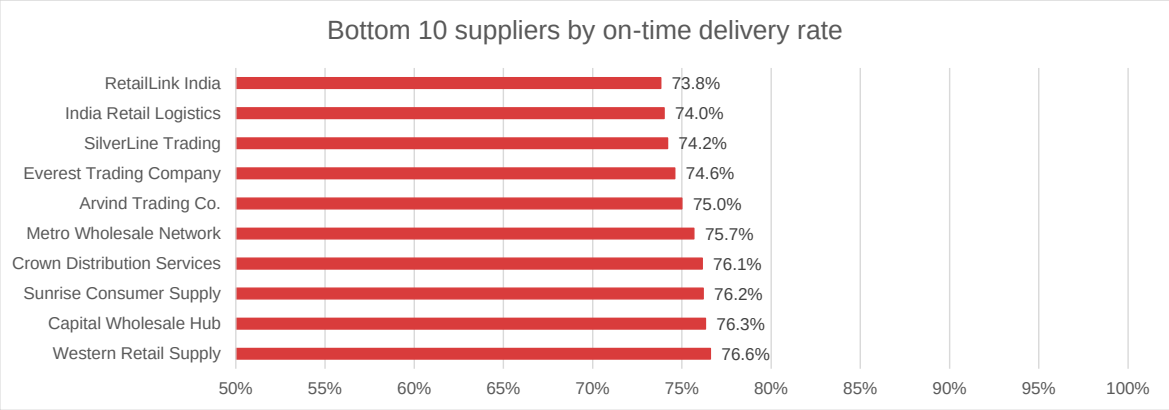


Figure 9: Bottom 10 suppliers by on-time delivery rate

Table 9: Supplier scorecard - five least and five most punctual suppliers

Supplier_Name	Received POs	On-Time %	Fill Rate %	Avg Days Late (late POs)	Avg Lead Time (days)
RetailLink India	42	73.8%	90.9%	3.5	12.9
India Retail Logistics	50	74.0%	92.4%	5.1	12.3
SilverLine Trading	31	74.2%	90.8%	4.1	12.1
Everest Trading Company	63	74.6%	91.1%	4.3	8.1
Arvind Trading Co.	52	75.0%	91.3%	5.2	8.3
...					
Apex Consumer Distributors	79	86.1%	91.8%	4.5	12.6
Rapid Retail Logistics	56	87.5%	92.1%	3.9	12.5
NorthPoint Consumer Supply	81	87.7%	92.2%	5.1	5.6
Vertex Wholesale Pvt. Ltd.	53	88.7%	91.0%	5.2	4.6
Mohan Enterprises	49	89.8%	91.1%	4.4	7.4
Grand Total	**2,465**	**80.6%**	**91.1%**	**4.6**	**8.5**

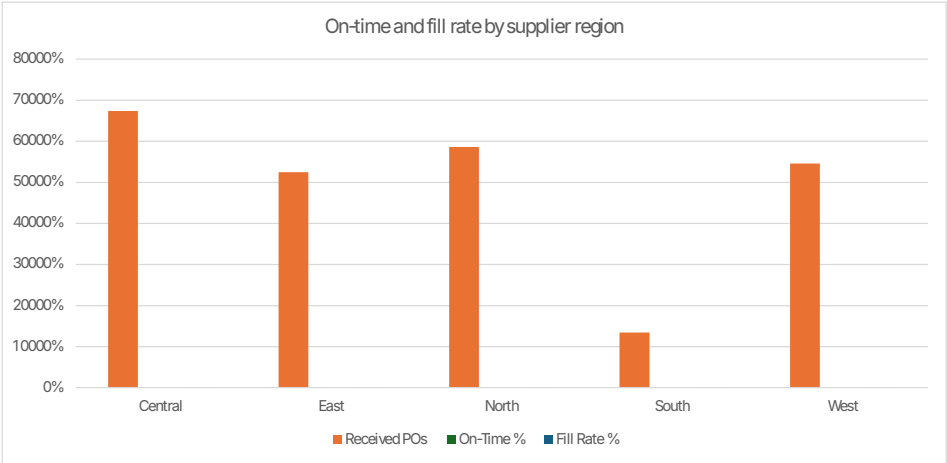


Figure 10: On-time and fill rate by supplier region

4.6 Where are inventory losses occurring? (Q6)

Approved negative stock adjustments removed 2,304 units worth ₹2.06 Cr at cost. Damage is the largest cause (₹0.91 Cr, 44%), followed by shrinkage (₹0.70 Cr, 34%); cycle-count and transfer corrections make up the rest. Losses follow unit value: Electronics accounts for ₹1.57 Cr (76%) despite a similar number of units lost as other categories. Delhi NCR records the highest losses (₹0.65 Cr, of which ₹0.37 Cr is damage). 40 adjustments are still awaiting approval (14 at Kolkata).

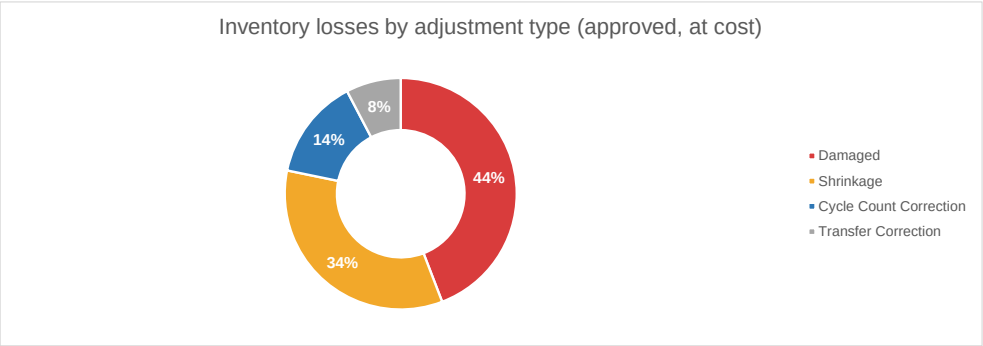


Figure 11: Share of approved inventory losses by adjustment type

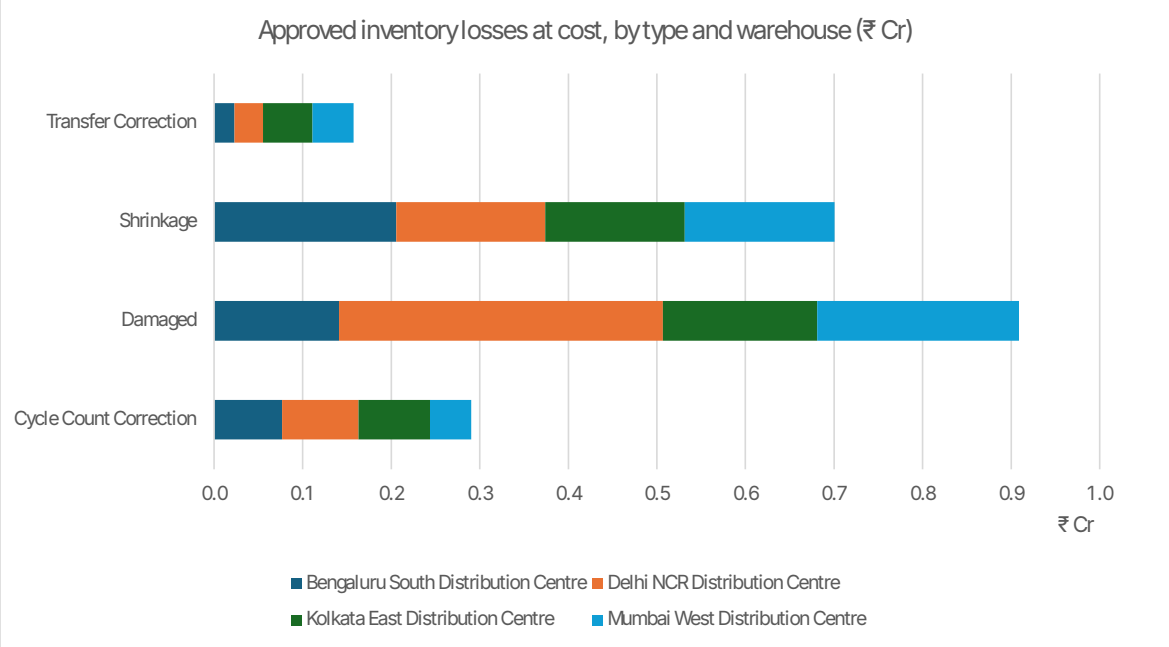


Figure 12: Approved inventory losses by adjustment type and warehouse (₹ Cr)

Table 10: Stock adjustments and losses by warehouse

Warehouse_Name	Adjustments	Units Lost (approved)	Loss Value (₹ Cr)	Net Adj. Value (₹ Cr)	Unapproved Adjustments
Delhi NCR Distribution Centre	319	665	₹0.65 Cr	₹-0.57 Cr	8
Mumbai West Distribution Centre	297	590	₹0.49 Cr	₹-0.33 Cr	8
Kolkata East Distribution Centre	290	491	₹0.47 Cr	₹-0.34 Cr	14
Bengaluru South Distribution Centre	285	558	₹0.45 Cr	₹-0.38 Cr	10
Grand Total	**1,191**	**2,304**	**₹2.06 Cr**	**₹-1.62 Cr**	**40**

5. Recommendations (Q7)

The actions below aim to release cash tied up in excess and slow-moving stock while protecting sales by fixing stockouts. They are ordered by urgency; each is tied to the evidence in Section 4.

Table 11: Prioritised action plan

Priority	Action	Evidence	Expected impact
1 - Now	Rebalance stock between warehouses before placing new POs: move units from excess pairs to stockout pairs of the same product (e.g. IdeaPad Slim 5 at Bengaluru).	308 excess vs 321 stockout pairs; ₹ 14.00 Cr above target	Fills stockouts with stock already owned; releases cash without new buying.
2 - Now	Replenish the 452 at-risk pairs in Table 3.4, starting with the 79 A-class pairs and the Delhi NCR and Mumbai West DCs.	321 stockouts + 131 reorder risk; Delhi 39.6% stockout rate	Protects revenue on the fastest-selling items; reduces lost sales (41,962 units historically).
3 - 30 days	Freeze or cut open/new POs for pairs above Target_Stock until they fall back to target; review targets for the top-value SKUs (IdeaPad Slim 5, 43-inch 4K TV, Galaxy M35 5G).	₹ 14.00 Cr above target; 12-24 months of cover on top SKUs	Stops excess from growing; gradually releases the above-target ₹ 14 Cr.
4 - 30-60 days	Clear slow movers: targeted website/app promotions, bundles or vendor returns for the 110 slow-moving pairs (Table 4.4), Fashion and Kolkata first.	₹ 4.25 Cr in 110 pairs with <10 units sold in 90 days	Converts dead stock to cash and frees space; avoids future write-offs.
5 - 60 days	Supplier scorecard and lead-time buffers: put the 5 suppliers at or below 75% on-time on improvement plans with penalties; shift volume to suppliers above 85%; add safety stock where suppliers are late.	On-time 80.6%, fill 91.1%; 4.6 days average delay	Fewer stockouts caused by late/short deliveries; more reliable reorder points.
6 - 60 days	Reduce handling damage and shrinkage: Electronics handling SOPs and secure storage, starting at Delhi NCR; monthly cycle counts on high-value SKUs; clear the 40 unapproved adjustments.	₹ 2.06 Cr losses; damage 44%; Electronics 76%	Lower losses and more accurate stock records for replenishment.
7 - Ongoing	Refresh this workbook monthly (Data > Refresh All) and track targets: stockout rate below 15%, above-target value below ₹ 7 Cr, supplier on-time above 90%, turnover above 3.0x.	Current: 32.1% stockout rate, ₹ 14.00 Cr above target, 80.6% on-time, 2.82x turnover	Makes inventory health visible to management every month.

6. KPIs to track and next steps

We recommend management reviews the Excel dashboard monthly after refreshing the data, and tracks the following targets:

Table 12: Targets for the monthly inventory review

KPI	Current (31-Aug-2026)	Target	Why it matters
Stockout rate (product-locations)	32.1%	Below 15%	Directly drives lost sales
Value above target stock	₹14.00 Cr	Below ₹7 Cr	Cash tied up in excess stock
Supplier on-time rate	80.6%	Above 90%	Late POs cause stockouts
Supplier fill rate	91.1%	Above 95%	Short shipments cause stockouts
Inventory turnover	2.82x	Above 3.0x	Efficiency of inventory investment

Next steps

- Week 1-2: agree the transfer list (excess to stockout pairs) and the PO freeze for items above target with the merchandising and supply chain teams.
- Week 2-4: replenish A-class stockout and reorder-risk items; launch promotions for the 110 slow-moving pairs.

- Month 2: share supplier scorecards and agree improvement plans; review reorder points and safety stock for late suppliers.
- Ongoing: refresh the workbook monthly (Data > Refresh All) and review the dashboard KPIs against the targets above.

Appendix A. Excel Inventory Health Dashboard

The interactive dashboard in the workbook (sheet "Dashboard") shows 12 KPI cards and 8 PivotCharts. Slicers for warehouse, category and ABC class filter every card and chart. It is shown here unfiltered.

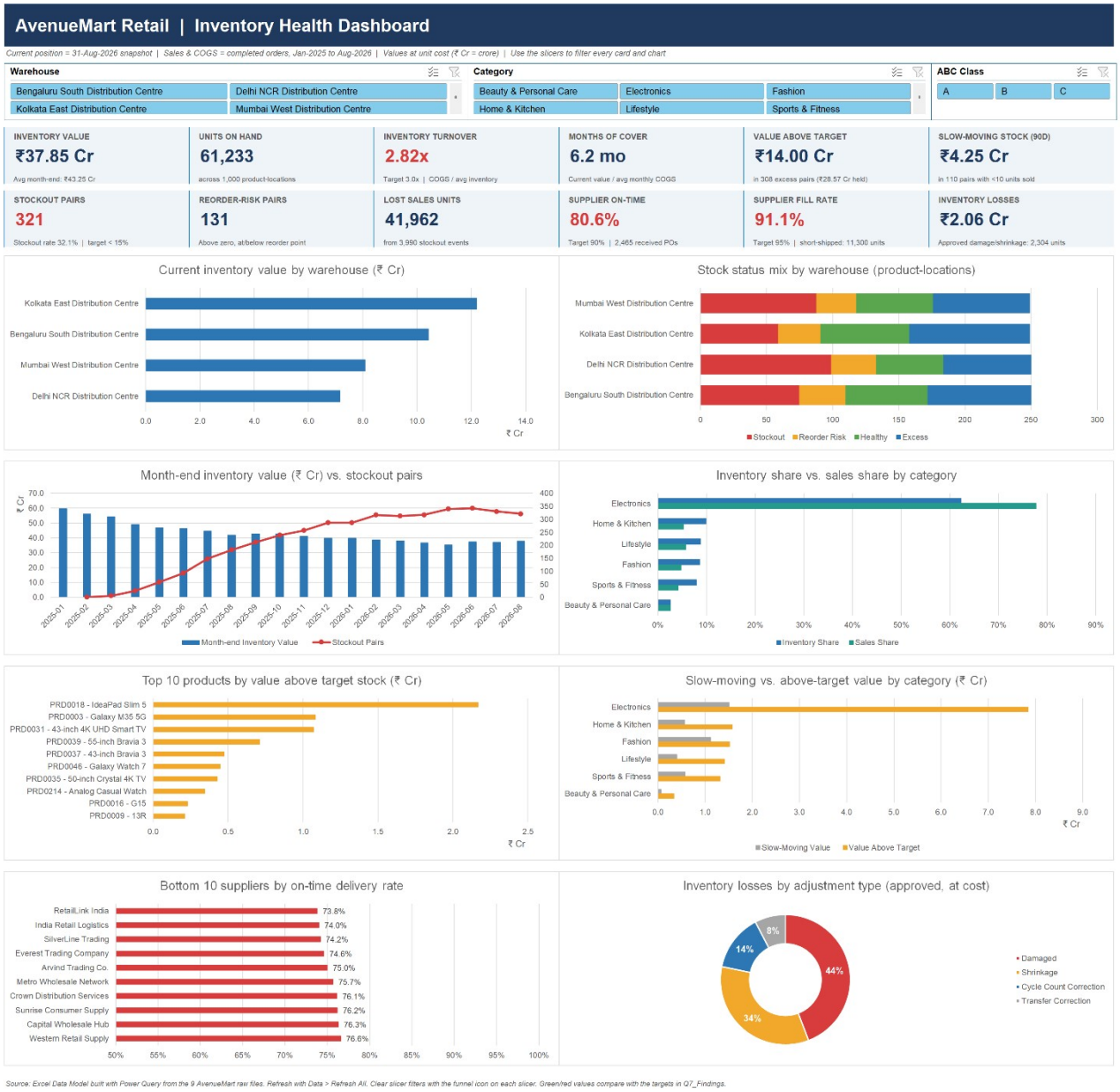


Figure 13: Inventory Health Dashboard in Excel (all warehouses, categories and ABC classes)

Appendix B. Data quality log

Table 13: Issues found and cleaning applied in Power Query

Table	Issues found	Cleaning applied in Power Query
dim_Products	Mixed-case IDs; categories and brands in 3 casings; ₹ symbols and commas in prices; stock policy in one field.	IDs standardised; best casing applied; numbers parsed; policy split into Reorder_Point, Safety_Stock, Target_Stock.
dim_Suppliers	Mixed-case IDs and status; profile combined as Region|Lead days|Payment terms.	IDs standardised; profile split into three columns; status in Title Case (38 Active, 2 Review).
dim_Warehouses	Duplicate WH004 row; location combined as City|Region; capacity "38,000" stored as text.	De-duplicated; location split; capacity converted to a number.
dim_Calendar	Clean daily calendar, 1-Jan-2025 to 31-Aug-2026.	Date typed; year, month, quarter and month-end fields rebuilt from the date.
fact_Inventory	Combined warehouse|product key; 7-part stock flow in one field; ₹ in values; 163 negative adjustments on zero stock.	Keys and flow split; value = closing units x unit cost; stock status recalculated; 163 rows flagged in Flow_Check.
fact_Sales	Discounts in 3 formats (10%, 10, 0.10); ₹ and commas in amounts; 484 cancelled lines.	Discount normalised; amounts parsed; cancelled lines excluded from every sales, demand and COGS measure.
fact_PurchaseOrders	Combined supplier|warehouse field; 10 POs without a received date (7 Open, 3 marked Partially Received).	Fields split; on-time flag, delay days and short quantity added; unreceived POs excluded from service KPIs.
fact_StockAdjustments	Combined key; type and approval flag casing; 40 unapproved adjustments.	Standardised; adjustment value at unit cost; only approved negative adjustments count as losses.
fact_StockoutEvents	Details combined as Days|Lost units|Lost value.	Split into three columns; lost units reconcile to the inventory snapshots (41,962).

Appendix C. Key DAX measures

Table 14: Selected DAX measures from the Excel Data Model

Measure	DAX formula
Latest Snapshot Date	`CALCULATE ( MAX ( fact_Inventory[Snapshot_Date] ), ALL ( fact_Inventory[Stock_Status] ), ALL ( dim_Products ), ALL ( dim_Warehouses ) )`
Current Inventory Value	`VAR d = [Latest Snapshot Date] RETURN CALCULATE ( SUM ( fact_Inventory[Inventory_Value] ), fact_Inventory[Snapshot_Date] = d )`
Current Stockout Pairs	`VAR d = [Latest Snapshot Date] RETURN CALCULATE ( COUNTROWS ( fact_Inventory ), fact_Inventory[Snapshot_Date] = d, fact_Inventory[Stock_Status] = "Stockout" )`
Above-Target Excess Value	`VAR d = [Latest Snapshot Date] RETURN CALCULATE ( SUM ( fact_Inventory[Excess_Value_Above_Target] ), fact_Inventory[Snapshot_Date] = d )`
Inventory Turnover	`DIVIDE ( [Completed COGS], [Avg Monthly Inventory Value] )`
Slow-Moving Pairs (90D)	`COUNTROWS ( FILTER ( CROSSJOIN ( VALUES ( dim_Warehouses[Warehouse_ID] ), VALUES ( dim_Products[Product_ID] ) ), [Current Inventory Units] > 0 && [Units Sold (Last 90D)] + 0 < 10 ) )`
Completed Sales	`CALCULATE ( SUM ( fact_Sales[Net_Sales] ), fact_Sales[Order_Status] = "Completed", fact_Sales[In_Study_Period] = "Yes" )`
Units Sold (Last 90D)	`VAR d = [Latest Snapshot Date] RETURN CALCULATE ( SUM ( fact_Sales[Quantity] ), fact_Sales[Order_Status] = "Completed", ALL ( dim_Calendar ), fact_Sales[Sale_Date] > d - 90, fact_Sales[Sale_Date] <= d )`
Supplier On-Time Rate %	`DIVIDE ( [On-Time Received POs], [Received POs] )`
Supplier Fill Rate %	`DIVIDE ( CALCULATE ( SUM ( fact_PurchaseOrders[Received_Qty] ), fact_PurchaseOrders[Is_Received] = 1 ), CALCULATE ( SUM ( fact_PurchaseOrders[Ordered_Qty] ), fact_PurchaseOrders[Is_Received] = 1 ) )`
Approved Negative Adjustment Value	`- CALCULATE ( SUM ( fact_StockAdjustments[Adjustment_Value] ), fact_StockAdjustments[Approved_Flag] = "Yes", fact_StockAdjustments[Adjustment_Qty] < 0 )`